FutureProof Financial

Equity Preservation Mortgage v14d (Optimised)

Product-Design Improvements

INTERNAL — FOR TEAM DISCUSSION

Running a range of product configurations through the model, this paper sets out the dials that give the borrower more income, FutureProof / lenders / funders more revenue, and less risk — and which of them are near-pure wins versus genuine trade-offs. It closes with one product assumption to confirm: how the principal is repaid.

Risk is shown two ways: the per-mortgage **probability of deficit (PoD)** — a balance-sheet measure, *not* a borrower loss rate — and the **reinsurance fair premium**, the discounted expected loss on the tail layer, which captures severity rather than just frequency. The reinsurance PoC quoted alongside is, by construction, 0.20 × PoD (the tail layer attaches at the worst-20%-of-deficits boundary).

Provenance. All figures are computed from the validated fast engine (epm_engine_v14d.py), 50,000 paths, seed 42. The engine ties out to Pavel's authoritative workbook and reads about 0.85pp conservative on PoD (base case: engine 9.2% vs workbook 8.4%). The *relative* improvements below are robust; the absolute levels are a slightly conservative read of the workbook. Built 2026-06-01.

1. What's the best we can do

Sweeping the product levers (annuity, payout term, FutureProof margin, profit share) at a reinsurance-PoC ceiling of 2% (investment-grade) and the current collar floor of 0.80, with borrower income prioritised, the standout is a single redesign that improves the borrower, FutureProof and the risk profile at once:

Configuration	Borrower annuity	PoD	Reins PoC	Reins premium	FP revenue	Lender NIM	Funder margin
Current (300K / 10yr / FP 0.50%)	$300K	9.2%	1.84%	$1,814	$936K	$237K	$678K
Recommended (350K / 25yr / FP 0.25%)	$350K	5.4%	1.08%	$897	$954K	$217K	$620K

Both rows use the current collar floor 0.80, so the comparison is like-for-like and does not depend on any change to the hedge. **Widening the floor to 0.75 (–25%) is the model optimum** and would take the recommended config to PoD 4.1% / reins premium $652 / FP $1,037K (+11%) — but it sits outside the current ±20% collar constraint and rests on skew-free pricing (see the caveat below), so it is treated here as upside, not the base recommendation.

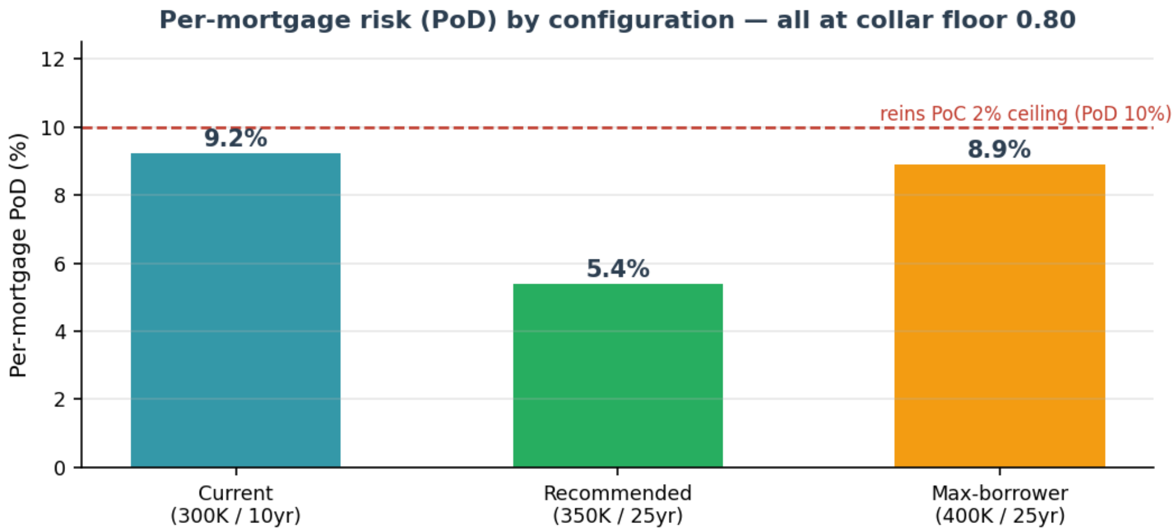
The recommended config (at floor 0.80) is a near-pure win:

- Borrower income **+17%** ($350K vs $300K annuity).
- Risk down sharply — PoD **9.2%** → **5.4%**, and the reinsurance fair premium –**51%** ($1,814 → $897), the severity-aware measure.
- FutureProof revenue **+2%** ($936K → $954K); at the wider floor 0.75 this rises to +11%.
- The only give-up: lender NIM and funder margin fall **~8%** (–8%) — a longer payout term means a lower average loan balance, and those margins are loan-based.

Pushing the borrower to $400K is feasible (+33%) but PoD rises to 8.9% and FutureProof revenue falls –13% ($814K), so it is a genuine choice rather than a free gain. Three levers drive the win:

- **Longer payout term (25yr)** draws the annuity slowly, keeping the loan lower for longer → less interest drag → lower risk and room for more annuity.
- **Lower FP margin (0.25%)** raises FP revenue — FutureProof takes 50% of the maturity surplus, so a smaller fee leaves a fatter account and FP's share of the bigger surplus more than compensates.
- **Keep profit-share at 10%** — taking profit earlier reduces the compounding base.

The collar floor is a separate lever — judge it on the premium, not the PoC. The reinsurance PoC (0.20×PoD) is frequency-only and cannot see what the collar actually changes, which is tail *severity*. On the severity-aware reinsurance premium, the current 0.80 floor over-pays: widening to 0.75 cuts the premium from $897 to $652 (–64% vs the current product). But this uses Black-Scholes pricing with *no volatility skew* — real –25% index puts carry a skew premium, which is likely why the ±20% / floor-0.80 constraint exists. Confirm the skew-adjusted hedge cost with the hedging desk before relaxing the constraint; floor 0.80 is the safe base.


Per-mortgage risk (PoD) by configuration — all at collar floor 0.80
Per-mortgage PoD (%)
reins PoC 2% ceiling (PoD 10%)
9.2%
5.4%
8.9%
0
2
4
6
8
10
12
Current
(300K / 10yr)
Recommended
(350K / 25yr)
Max-borrower
(400K / 25yr)

2. A new risk dial — the "ratchet"

A naive calendar glide (de-risking the investment toward cash on a fixed schedule) *raises* risk here — the account must keep growing against the debt, so de-risking every path just sacrifices the growth. But a **state-dependent ratchet** is different: it de-risks **only the winners**. Once the account is safely above the loan (balance ≥ loan × (1+buffer)), the surplus cushion is moved to cash and locked in, keeping the obligation amount in equity.

De-risking rule	PoD	Reins premium	Mean surplus	FP revenue
None (100% equity, current)	9.2%	$1,814	$1.11M	$936K
Ratchet (lock above loan ×1.1)	7.6%	$1,466	$0.80M	$748K
Naive calendar glide (→50% by yr20)	14.6%	$2,804	$0.73M	$738K

The ratchet cuts PoD and the reinsurance premium by locking in gains; it costs mean surplus and FP revenue, so it is a **risk/return dial**, not a free lunch — useful if you want to reach a tighter reinsurance target than the levers in Section 1 achieve. The naive calendar glide is dominated on every axis and should not be used.

3. One assumption to confirm — how the principal is repaid

Every figure above assumes the model's current treatment of the loan principal: it is drawn to its $1.2M peak and then **held flat**, with interest paid each year out of the investment account and the principal repaid as a single **lump at maturity**. This is reverse-mortgage-like and consistent with the fact that the EPM borrower makes no repayments — they receive an annuity. We believe this is correct, but the "P&I" label makes it sound as though the balance should amortise, so it is worth one explicit confirmation.

Why it matters: if instead the principal were meant to amortise to zero over the term, the only source for those repayments is the investment account itself — draining the growth the product relies on. On the same engine the base-case PoD rises from **9.2% to 12.0%** (workbook-equivalent base ~8.4%), the current product's reinsurance PoC would breach the 2% target, and the recommended design above would need re-running. So please confirm: **flat principal with a lump repayment at maturity — yes?**

4. Recommendation

- **Adopt the recommended design** (350K annuity / 25yr payout / FP margin at its 0.25% floor, current collar 0.80): borrower +17%, FutureProof +2%, PoD 9.2%→5.4%, reinsurance premium –51%. This is robust and does not depend on the hedge.
- **Decide whether to relax the collar floor to 0.75** with the hedging desk. The model says it improves every axis (FP +11%, PoD to 4.1%), but on skew-free pricing and outside the current ±20% constraint — verify the real hedge cost first.
- **Don't add a calendar glide path**, but keep the **state-dependent ratchet** in reserve as a risk dial if you want to reach a tighter reinsurance target.
- **Decide the borrower-vs-stakeholder balance** for the annuity level: $350K is a near-pure win; $400K maximises borrower income (+33%) but costs FutureProof revenue (–13%) and ~8% of lender/funder margin.
- **Confirm the flat-principal assumption** (Section 3).